

BALA VIGNESH R

Portfolio | [LinkedIn](#) | [GitHub](#) | [Leetcode](#)

Location: Tamil Nadu, India

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TECHNICAL SKILLS

Languages : Go, Java, C, Python, C++, JavaScript, SQL

Frameworks : Spring Boot

Technologies : REST APIs, WebSocket, Git, GitHub

Databases : PostgreSQL, SQLite

INTERNSHIP

Backend Intern

Jun 2025 – Jul 2025

Aziro Technologies India Pvt. Ltd. (HPE Project)

India

- Completed one-month backend internship focused on Golang development.
- Developed backend modules using Go under the guidance of the development team.
- Collaborated with engineers to understand backend architecture and industry development practices.

EDUCATION

Vellore Institute of Technology, Chennai

Tamil Nadu, India

B.Tech in Computer Science (CGPA: 8.13)

2023 – 2027

Aristo Public School, Cuddalore

Tamil Nadu, India

Higher Secondary (Class XII) – 86.4%

2023

PROJECTS

AIGIS: Autonomous Integrity Guard for Intercloud Storage Python | SQLite | AES-256 | Shamir Secret Sharing | Drive API

- Developed a decentralized **Zero-Trust cloud storage system** using AES-256 GCM encryption, Shamir's Secret Sharing and Reed-Solomon Erasure Coding.
- Distributed encrypted file shards across Google Drive, Dropbox and Mega.nz with an autonomous Moving Target Defense (MTD) mechanism.
- Implemented TOTP-based authentication and demonstrated **100% data recovery** after complete failure of one cloud provider (K=2, N=3 threshold).

ESP32-Based Gas Follower Robot with Web Dashboard

ESP32 | C++ | WebSocket | HTML | CSS | JavaScript

- Built an autonomous gas leak localization robot using an ESP32 microcontroller with a three-sensor MQ gas array.
- Developed a real-time WebSocket dashboard supporting live telemetry, 2D radar visualization and manual W-A-S-D teleoperation.
- Achieved an average communication latency of **42 ms** across 1,000 test runs and implemented autonomous danger-state navigation.

Smart Network Management and Intrusion Detection System (SNMIDS)

Node.js | Express | Chart.js

- Developed a web-based network monitoring and intrusion detection system for small and medium enterprises.
- Implemented real-time traffic visualization, rule-based and anomaly-based intrusion detection with mitigation alerts.
- Currently improving monitoring accuracy, server portability and overall system security.